

Hugon Lee

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RESEARCH INTEREST

Micromechanics of composites; physics-informed machine learning; data-driven design and optimization; agentic AI

EDUCATION

Korea Advanced Institute of Science and Technology (KAIST)		Daejeon, Republic of Korea
Ph.D. Mechanical Engineering (Expected Aug. 2026)		Advisor: Seunghwa Ryu
Feb. 2022- Aug. 2026 (planned)	• Thesis: “Interface and interphase effects in stiff and soft composites: micromechanics and physics-informed machine learning” • GPA (M.S.-Ph.D., cumulative): 4.04/4.30	
M.S. Mechanical Engineering		Advisor: Seong Su Kim
Feb. 2020- Feb. 2022	• Thesis: “(A) study on permeability prediction of fiber-reinforced composite materials based on the representative volume element”	
B.S. Mechanical Engineering		
Mar. 2014- Feb. 2020	• <i>Dean's list</i>, College of Engineering, 2018 Fall semester • <i>Academic Excellence Award</i>, Mechanical Engineering, 2018 Fall and 2019 Spring semesters	

RESEARCH EXPERIENCE

Oct. 2022- Present	Mechanical behavior of composites with interface/interphase effects • Developed and validated interface- and interphase-enriched mean-field homogenization models for matrix-inclusion composites, including core-shell systems • Conducted comparative assessment of mean-field approaches against full-field FE benchmarks across varying material contrast • Extended mean-field stress-state predictions to coupled mechanoluminescent response modeling in composite systems • Applied the mean-field framework to injection-molded short fiber composites and constructed multi-fidelity data-driven surrogates for effective property prediction • Outcome: <i>Multiscale Sci. Eng.</i> (2023, Review); <i>Addit. Manuf.</i> (2024); <i>J. Phys. Condens. Matter</i> (2024); <i>manuscripts in preparation</i>
Mar. 2022- Dec. 2024	Simulation-driven design optimization under manufacturing constraints • Developed scripted FE simulation workflows and Bayesian optimization pipelines for process design and structural optimization of injection-molded components • Outcome: <i>Adv. Theory Sim.</i> (2024a,b); <i>J. Manuf. Process.</i> (2025); <i>Adv. Intell. Disco.</i> (2025, Perspective)

Peer-reviewed

13. Kim, H., **Lee, H.**, Park, M. and Ryu, S.* “Morphology-, noise-, and resolution-robust ultrasound elasticity imaging with Fourier neural operator.” *Computer Methods and Programs in Biomedicine* 283, no. 15 (2026): 109432.
<https://doi.org/10.1016/j.cmpb.2026.109432>
12. Lee, I., Chae, H., Park, J., Shin, J., **Lee, H.*** and Ryu, S.* “Comparative assessment of composition- and structure-based surrogate models across 2D materials databases.” *Physical Chemistry Chemical Physics* 28, no.16 (2026): 10119–10130.
<https://doi.org/10.1039/D5CP04814A>
11. **Lee, H.†**, Moon, H.†, Lee, J.† and Ryu, S.* “Toward knowledge-guided AI for inverse design in manufacturing: A perspective on domain, physics, and human-AI synergy.” *Advanced Intelligent Discovery* (2025): e202500107.
<https://doi.org/10.1002/aidi.202500107> (Perspective article)
10. Cho, H., **Lee, H.** and Ryu, S.* “Sequential multi-objective Bayesian optimization of air-cooled battery thermal management system with spoiler integration.” *Journal of Energy Storage* 114 (2025): 115586.
<https://doi.org/10.1016/j.est.2025.115586>
9. **Lee, H.†**, Yeo, J.†, Kong, K., Myeong, D., Jang, D., Lee, J., Choi, H., Kim, N.* and Ryu, S.* “Bayesian optimization of tailgate rib structures enhancing structural stiffness under manufacturing constraints of injection molding.” *Journal of Manufacturing Processes* 134 (2025): 739-748.
<https://doi.org/10.1016/j.jmapro.2024.12.064>
8. **Lee, H.**, Yeo, J., Yu, J., Moon, H. and Ryu, S.* “Mean-field homogenization of liquid metal elastomer composites: Comparative study and exact dilute solution of core-shell inclusion.” *Journal of Physics: Condensed Matter* 36, no. 50 (2024): 505702.
<https://doi.org/10.1088/1361-648X/ad765e>
7. **Lee, H.†**, Lee, M.†, Jung, J., Lee, I.* and Ryu, S.* “Enhancing injection molding optimization for SFRPs through multi-fidelity data-driven approaches incorporating prior information in limited data environments.” *Advanced Theory and Simulations* 7, no. 8 (2024b): 2400130.
<https://doi.org/10.1002/adts.202400130>
6. Jo, J.†, Park, M.†, Kang, S.†, **Lee, H.†**, Gu, C.-Y., Kim, T.-S. and Ryu, S.* “Data-driven prediction of strain fields in auxetic structures and non-contact validation with mechanoluminescence for structural health monitoring.” *International Journal of AI for Materials and Design* 1, no. 2 (2024): 48-60.
<https://doi.org/10.36922/ijamd.3539>
5. Jung, J., Park, K., **Lee, H.**, Cho, B. and Ryu, S.* “Comparative study of multi-objective Bayesian optimization and NSGA-III based approaches for injection molding process.” *Advanced Theory and Simulations* 7, no. 7 (2024a): 2400135.
<https://doi.org/10.1002/adts.202400135>
4. Jo, J.†, Park, K.†, Song, H.†, **Lee, H.** and Ryu, S.* “Innovative 3D printing of mechanoluminescent composites: Vat photopolymerization meets machine learning.” *Additive Manufacturing* 90 (2024): 104324.
<https://doi.org/10.1016/j.addma.2024.104324>
3. Park, D.†, Lee, J.†, **Lee, H.**, Gu, G. X. and Ryu, S.* “Deep generative spatiotemporal learning for integrating fracture mechanics in composite materials: inverse design, discovery, and optimization.” *Materials Horizons* 11, no. 13 (2024): 3048-3065.
<https://doi.org/10.1039/D4MH00337C>

2. **Lee, H.**, Lee, S.* and Ryu, S.* “Advancements and challenges of micromechanics-based homogenization for the short fiber reinforced composites.” *Multiscale Science and Engineering* 5, no. 3 (2023): 133-146.
<https://doi.org/10.1007/s42493-024-00100-2> (Review article)
1. Lee, J., Park, D., Lee, M., **Lee, H.**, Park, K., Lee, I. and Ryu, S.* “Machine learning-based inverse design methods considering data characteristics and design space size in materials design and manufacturing: a review.” *Materials Horizons* 10, no. 12 (2023): 5436-5456.
<https://doi.org/10.1039/D3MH00039G> (Review article)

SELECTED ORAL PRESENTATIONS

†: equal contribution; *: corresponding author

5. **Lee, H.** and Ryu, S.* “Multi-fidelity surrogate for mechanical properties of short fiber reinforced plastics: leveraging information from theory to experiment.” *The 9th Asian Pacific Congress on Computational Mechanics/The 7th Australasian Conference on Computational Mechanics (APCOM-ACCM 2025)*, Brisbane, Australia, Dec. 2025.
4. **Lee, H.** and Ryu, S.* “Micromechanical analysis on interphase effects of liquid metal elastomer composites.” *The 2024 Conference of the Korean Society of Mechanical Engineers (KSME 2024)*, Jeju, Republic of Korea, Nov. 2024.
3. **Lee, H.**†, Jo, J.† and Ryu, S.* “Numerical implementation of digital light processing 3D printing process using Abaqus.” *The 26th International Congress of Theoretical and Applied Mechanics (ICTAM 2024)*, Daegu, Republic of Korea, Aug. 2024.
2. **Lee, H.**, Jung, J. and Ryu, S.* “Data-driven optimization of injection molding process parameters: comparisons of parts with varying complexity.” *The 2023 Spring Conference of the Korean Society of Mechanical Engineers (KSME), CAE & Applied Mechanics Division*, Busan, Republic of Korea, May 2023.
1. **Lee, H.** and Kim, S. S.* “A study on the optimal curing process of filament wound tow-preg laminated composites through multi-scale modeling.” *The 24th International Conference on Composite Structures (ICCS24)*, Virtual, Porto, Portugal, Jun. 2021.

TEACHING / MENTORING EXPERIENCES

Academic talks	2026 winter AI class of Korean Society for Composite Materials (KSCM), Republic of Korea (Feb. 2026) - Lecture and hands-on tutorial on LLM usage and tool-calling for homogenization workflows Invited talk, Prof. Patrick C. Lee group @ University of Toronto, Canada (Nov. 2023) - Delivered introduction to multi-fidelity machine learning and their utilization for experimental data
Student mentoring	Inhyo Lee (Jan. 2025 – Oct. 2025): - Project: research design & manuscript preparation; submitted to <i>Physical Chemistry Chemical Physics</i> Sangwoo Jung (Jun. 2024 – Oct. 2025): - Master’s research: modeling of DLP additive manufacturing process Hanbin Cho (Aug. 2023 – Aug. 2024): - Master’s research: numerical analysis & Bayesian optimization; published in <i>Journal of Energy Storage</i> Hyeonseong Jo (Feb. 2021 – Feb. 2022): - Master’s research: experimental design, permeability analysis, numerical modeling

PROFESSIONAL SERVICES

Journal Reviewer: Nature Communications, Mechanical Systems and Signal Processing, Applied Composite Materials, Progress in Additive Manufacturing, Journal of Thermal Analysis and Calorimetry, and Discover Mechanical Engineering

TECHNICAL SKILLS

Programming: Python (numerical analysis, tensor operations, optimization workflows, machine learning), Fortran (for Abaqus user subroutines), MATLAB (numerical analysis, tensor operations)

Computational analysis: Abaqus (scripting-assisted workflows, use/modification of user subroutines), COMSOL, HyperWorks (with scripting; TCL language), Moldflow (with scripting; PyMoldflow package)

Modeling: Micromechanics/homogenization (mean-field, computational), surrogate modeling (multi-fidelity), Bayesian optimization, process modeling of polymers/polymer composites (including DLP additive manufacturing)

Workflow: Script-based simulation workflows, parametric studies, automated pre-/post-processing, numerical benchmarking, machine learning (GPy, PyTorch), scientific visualization (Matplotlib, PyVista, Illustrator)

Experimental (design & validation): Experimental design for (composite) material evaluation and process characterization, definition of loading and measurement protocols, simulation-experiment comparison

REFERENCES

Dr. Seunghwa Ryu (Ph.D. advisor)

Professor, Department of Mechanical Engineering, Korea Advanced Institute of Science and Technology

Dr. Patrick C. Lee

Associate Professor, Department of Mechanical and Industrial Engineering, University of Toronto

Dr. Sangryun Lee

Assistant Professor, Department of Mechanical and Biomedical Engineering, Ewha Womans University